

Image Classification of Vehicle Brands using Deep Learning.

Subject: BIAI

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Overview

The main goal of the project is to develop, train, and evaluate an advanced deep learning model capable of precisely classifying images of cars into specific brands.

The development of this solution will enable the automation of vehicle recognition and brand categorization, demonstrating the effectiveness of advanced neural networks in visual data analysis.

Method of Implementation

- **Data Sources:** Publicly available image datasets from Kaggle
- **Programming Language:** Python 3.x
- **Core Frameworks:** PyTorch, TensorFlow
- **Libraries & Solutions:** timm, TorchMetrics, pytorch-grad-cam, PIL, torchvision, numpy, OpenCV
- **Environment:** Google Colab

Datasets

- **Car Brands Images - 100 Images Of Top 50 Car Brands:** A dataset consisting of 4,500 car images divided into 50 folders for each brand.
- **Cars Collection Dataset - Collection of over 60k images of cars:** A dataset consisting of 60,000 car images across 55 brands.

Examples of data



Current progress

- PyTorch CNN Model
- MobileNetV2 Model
- ResNet50 + TripleLoss with small dataset
- ResNet50 + TripleLoss with large dataset

CNN PyTorch Model

- **Architecture and modifications description:**
- **Architecture built from scratch:** Developed entirely in raw PyTorch.
- **Feature extraction:** Utilizes 3 convolutional layers for hierarchical feature extraction.
- **Classifier:** Based on fully connected layers (Linear).

Training Results

As a baseline for the entire project, a simple, three-layer convolutional neural network was designed. The objective was to evaluate the performance of a basic architecture on this task without utilizing external knowledge in the form of pre-trained weights. Training on a small dataset resulted in merely 9.8% accuracy on the validation set.

```
Wynik uczenia: 87.6% | Wynik na nowych zdjęciach: 8.1%  
Epoka 4/6: 100%|██████████| 230/230 [00:10<00:00, 22.94it/s]  
Wynik uczenia: 37.6% | Wynik na nowych zdjęciach: 8.2%  
Epoka 5/6: 100%|██████████| 230/230 [00:09<00:00, 23.53it/s]  
Wynik uczenia: 89.9% | Wynik na nowych zdjęciach: 9.0%  
Epoka 6/6: 100%|██████████| 230/230 [00:09<00:00, 24.53it/s]  
Wynik uczenia: 98.0% | Wynik na nowych zdjęciach: 9.8%
```


Confusion Matrix

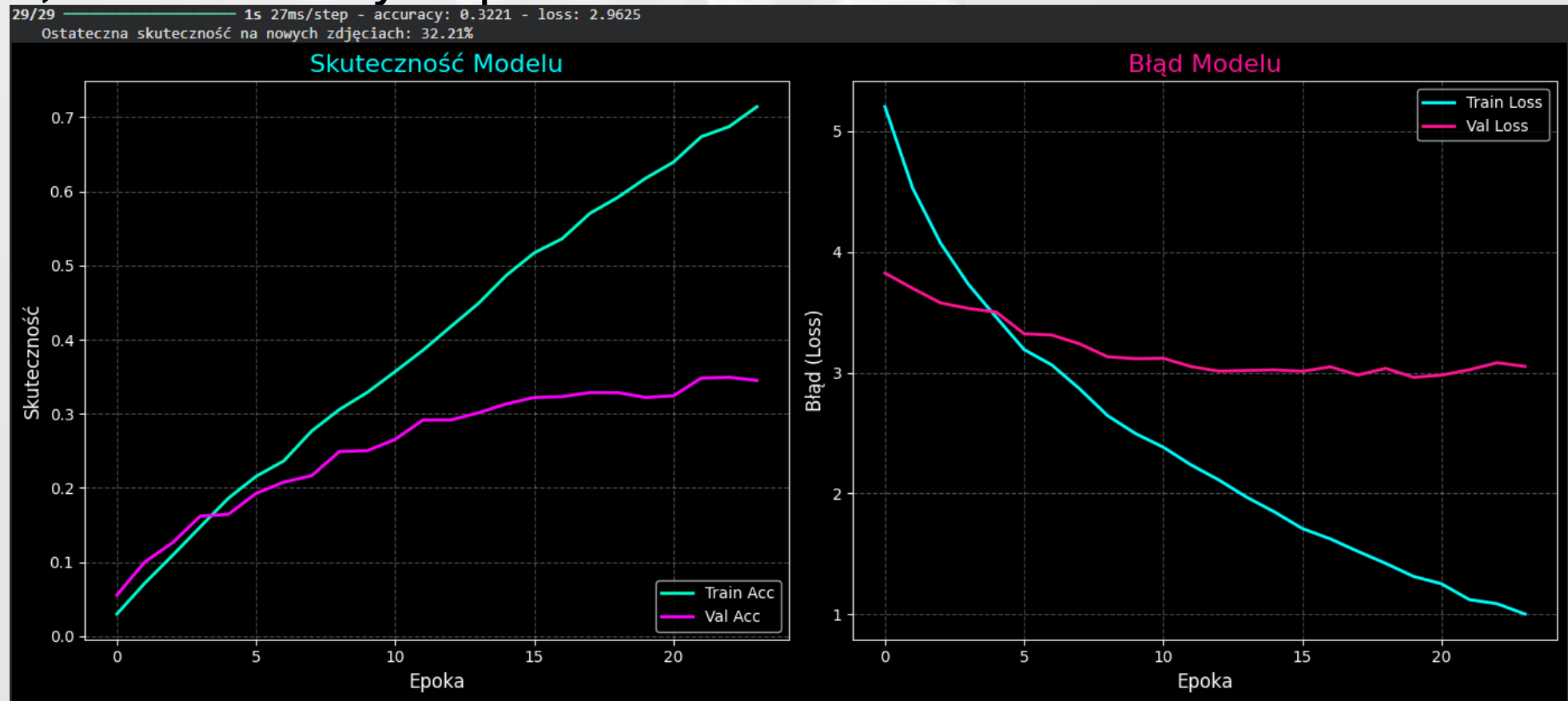
[illegible]

MobileNetV2 Model

- **Architecture and modifications description:**
- Applied the MobileNetV2 model (pre-trained on ImageNet) as a feature extractor instead of building a network from scratch.
- Implemented real-time data streaming (tf.data), which eliminated memory issues when processing the large image dataset.
- Fine-Tuning Technique: Partially "unfroze" the weights of the model's final block to better adapt it to the specific task of classifying 50 car brands.

Training Results

In the third stage of the project, the approach and tools were changed to evaluate the effectiveness of Transfer Learning. The MobileNetV2 architecture was selected for this purpose. As demonstrated by the charts and the results reaching approximately 32%, the accuracy improved.



ResNet50 + TripleLoss with small dataset

- **Architecture and modifications description:**
- Deep Architecture: Transitioned to the ResNet50 model utilizing pre-trained ImageNet weights (Transfer Learning).
- Dual Loss Function: Implemented a combined loss approach using CrossEntropyLoss (for classification) and CosineEmbeddingLoss (acting as Triplet Loss for feature clustering).
- Dataset: Remained unchanged (small dataset).

Training Results

Because simple networks proved insufficient, the ResNet50 architecture was implemented in the next stage. Rather than relying on standard classification, the network was trained to analyze fine details using a Triplet Loss function. As a result, the model achieved an impressive 99.1% accuracy on the validation set. However, when it was decided to test this model on random car images sourced from the internet to evaluate its real-world performance, the results were unfortunately not perfect, and some of the predictions were incorrect.

```
--- Epoka 14/15 ---
Trening: 100%|██████████| 230/230 [03:29<00:00, 1.10it/s]
Sprawdzanie: 100%|██████████| 29/29 [00:07<00:00, 3.88it/s]
Skuteczność (Z mat. uczył się): 99.4%
Skuteczność (Z mat. nowego): 99.3%
Zapisano nowy najlepszy model.

--- Epoka 15/15 ---
Trening: 100%|██████████| 230/230 [03:29<00:00, 1.10it/s]
Sprawdzanie: 100%|██████████| 29/29 [00:07<00:00, 3.87it/s]Skuteczność (Z mat. uczył się): 99.5%
Skuteczność (Z mat. nowego): 99.1%

Trening zakończony. Najlepszy wynik na walidacji: 99.3%
```

Confusion Matrix

Macierz Pomyłek

[illegible]

Przeiwdywanania siec

ResNet50 + TripleLoss with large dataset

- **Architecture and modifications description:**
- Data Augmentation/Expansion: Combined datasets to create a comprehensive database of >60,000 images (across 55 brands).
- Algorithmic Optimization: Reduced search time for mining pairs in Triplet Loss.

Training Results

During the traditional process of mining positive and negative samples for the Triplet Loss function, the CPU had to iterate through 60,000 files at each training step. This resulted in a single epoch taking approximately 4.5 hours, making the training process completely unfeasible. Therefore, an algorithmic optimization was designed. During the data loading stage, a dictionary was created to pre-group all file paths, which significantly reduced the computational complexity and resolved the bottleneck.

```
--- Epoka 3/15 ---  
Uczenie sieci: 100%|██████████| 3454/3454 [52:54<00:00, 1.09it/s]  
Walidacja: 100%|██████████| 432/432 [01:00<00:00, 7.14it/s]  
Dokładność (Trening): 81.14%  
Dokładność (Walidacja): 82.94%  
Znaleziono lepsze parametry Zapisuje 'najlepszy_model_pytorch.pth'  
  
--- Epoka 4/15 ---  
Uczenie sieci: 100%|██████████| 3454/3454 [52:57<00:00, 1.09it/s]  
Walidacja: 100%|██████████| 432/432 [01:00<00:00, 7.20it/s]  
Dokładność (Trening): 88.59%  
Dokładność (Walidacja): 85.88%  
Znaleziono lepsze parametry Zapisuje 'najlepszy_model_pytorch.pth'
```


Conclusions and development plan

- Simple CNN models (built from scratch) demonstrate a limited capacity for extracting complex visual features; in contrast, architectures like ResNet50 utilizing Transfer Learning drastically improve classification accuracy.
- The volume and diversity of data are crucial—eliminating the Domain Shift phenomenon was only possible after expanding the database to 60,000 images and applying Triplet Loss optimization.
- Vision Transformer (ViT) Implementation: Further stages of the project involve testing an architecture based on the Self-Attention mechanism as a modern alternative to traditional CNN networks.